Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/24/2020
PI Name:	O'Rourke, Aubrie Ph.D.		
Project Title:	Virulence and Drug Resistance of Burkholderia Species Isolated from ISS Potable Water Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	aubrie.e.orourke@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Mail Code DNPS		
PI Address 2:			
PI Web Page:			
City:	Cape Canaveral	**State:**	FL
Zip Code:	32899	**Congressional District:**	8
Comments:	Updated contact information. PI moved to Kennedy Space Center after the end date of this project. Dr. O'Rourke was formerly with J Craig Venter Institute (Ed, 2/15/23).		
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-MOBE. Appendix B: Post-Doctoral Fellowships to Study the Microbiome of the ISS as a Built Environment
Start Date:	06/06/2017	**End Date:**	09/01/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/1/2019 per NSSC information (Ed., 1/24/2020)		
Key Personnel Changes/Previous PI:	NOTE: Mentor change to Chris L. Dupont, Ph.D. in May 2018.		
COI Name (Institution):	Dupont, Chris L. Ph.D. (MENTOR: J. Craig Venter Institute)		
Grant/Contract No.:	80NSSC17K0035		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by NASA as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. The disease-causing organisms Burkholderia multivorans, B. cepacia, genomovars of the B. cepacia complex (BCC), as well as an unclassified species of Burkholderia, and its relatives Ralstonia pickettii and R. insidiosa account for 60% of the bacterial isolates obtained from the filtrated water system on board the International Space Station (ISS). Members of these genera can survive long periods in a planktonic state in distilled water or as biofilms within an infected individual. They are also natively resistant to several antibiotics leaving treatment options complex. For BCC organisms, this complexity is increased by the phenomenon of bacterial persistence where a genetically identical subpopulation of the bacteria sustains under high antibiotic challenge by a reduction in metabolism and other unknown mechanisms. A healthy astronaut could easily clear a BCC or Ralstonia sp. infection; however, after prolonged spaceflight, a potentially immunocompromised astronaut with repeated exposure to these organisms could be at risk for disease. To understand the mechanisms that would contribute to altered virulence in ISS BCC and Ralstonia species, the genomes of 67 isolates from 22 missions will be sequenced and compared to their respective ATCC BCC or Ralstonia strain. The potential for increased virulence will be determined by the presence of genes clustered as pathogenicity islands in the respective genome. To establish if the strains are becoming more resistant to antibiotics, a minimum inhibitory concentration (MIC) will be determined for each isolate using six antibiotics currently used to treat the infections and then compared to the respective ATCC control strain. Finally, a transposon library will be generated for the most virulent, resistant, tolerant, or persistent strain, and this library will be cultured in distilled water to simulate conditions on the ISS. Here we will identify genes which aid in the survivorship of BCC and Ralstonia strains in the water system of the ISS. The knowledge gained from this genomic sequencing and Tnseq study will allow us to assess any increased virulence or infection potential, as well as help to make antibiotic recommendations, predict target genes where inhibition by adjuvants will enhance antibiotic therapy, or present novel water treatment options to eradicate the BCC/Ralstonia strains from the potable water of the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of the project # 80NSSC17K0035 entitled, “Virulence and drug resistance of Burkholderia species isolated from ISS potable water systems” is to use “omic” studies to characterize the Burkholderia cepacia complex (BCC) and Ralstonia species that have been isolated from the potable water dispenser (PWD) of International Space Station (ISS) across 22 missions. The PWD is the source of all consumed water aboard the ISS; therefore, the surveillance of these bacterial species, which are known to be opportunistic pathogens, is relevant to ensure crew health. These genera are commonly found as soil organisms with the ability to survive in water. As a result, they are often found as contaminants in treated water supplies. These bacterial isolates cultured from the ISS PWD have originated from Earth. The effects upon them, as the result of exposure to microgravity, lowered shear forces, and increased radiation in the Earth’s upper atmosphere where the ISS orbits, are unknown. Our work benefits life on Earth by way of planetary protection, a facet of NASA since the start of the Apollo missions. The goal of planetary protection is to prevent any forward contamination to other celestial bodies as well as to prevent any back contamination to Earth. Our work will shed light on new ways to contain or treat these genera isolated form the ISS PWD, as well as to provide a better understanding of the structure and plasticity within the genomes of these space-adapted terrestrial bacteria.
	FINAL REPORTING JANUARY 2020 The overall goal of the project # 80NSSC17K0035 is to use “omic” studies to characterize the Burkholderia cepacia complex (BCC) and Ralstonia species that have been isolated from the potable water dispenser (PWD) of International Space Station (ISS). The PWD is the source of all consumed water aboard the ISS; therefore, the surveillance of these species, which are known to be opportunistic pathogens, is relevant to ensure crew health. The isolates were evaluated for any increase in virulence or infection potential and assayed for antibiotic resistance to address. The genomes and phenotypic results for the ISS BCC isolates have now been written up and the manuscript has been accepted by PLOS One as of December 16, 2019. In Year 2, a transposon library was successfully constructed in an ISS Burkholderia contaminans isolate. However, we were unable to run the experiment due to the elemental iodine selected for treatment being a controlled pre-cursor substance in the state of California; unfortunately, the experiment has been on hold due to the inability to procure iodine. Overall our research shows that within each species, the 19 B. cepacia and 5 B. contaminans, recovered from the ISS PWD over roughly a 5-year timespan, were highly similar on a whole genome scale, suggesting each population (here meaning one for each species) that was sampled may have stemmed from two distinct founding strains. This is supported by all isolates demonstrating greater than 99% ANI with 95-99.9% alignment in all cases, and sharing a core pangenome of greater than 90% of identified gene clusters (within species). The vast majority of differences in gene-content that were identified in the pangenome were localized on putative plasmids. Physiologically, including antibiotic sensitivities, the ISS-derived isolates were overall generally similar to each other, as well as to the reference type-strains analyzed. In cases where ISS-derived isolates did vary, no strong correlations were found between physiology and either single-nucleotide variant (SNV) distances or date of isolation (though it is also import to keep in mind date of isolation does not necessarily indicate any information about how long an isolate had been aboard the ISS). While it is interesting the 5 recovered B. contaminans isolates were all isolated in a span of time over 2013 and 2014 during which no B. cepacia were isolated, we cannot speak at this time to what may have caused this (if anything). That aside, our analysis here suggests these two species’ populations sampled by way of isolation seem to be highly similar genetically regardless of date of isolation, with the majority of variation that does exist being maintained among putative plasmids. Based on our phenotypic evaluation of the ability of these isolates to lyse or intracellularize within macrophage, in comparison to the reference type-strains, it seems likely that the two populations of Burkholderia present in the ISS PWS are not more virulent than those that might be encountered on planet.

Task Progress:	Completed Publications 1. Aubrie O'Rourke, Michael D. Lee, William C. Nierman, R. Craig Everroad, Chris L. Dupont. Genomic and phenotypic characterization of Burkholderia isolates from the potable water system of the International Space Station. PLOSONE. Accepted and in final stages before publication. 2. Aubrie O'Rourke, Angela Zoumplis, Paul Wilburn, Michael D. Lee, Zhi Lee, Marissa Vecina, Kysha Mercader. Following the Astrobiology Roadmap: Origins, Habitability and Future Exploration. Astrobiology for Caister Academic Press and as a Focus Issue of the journal Current Issues in Molecular Biology. Accepted and published. ANNUAL REPORTING APRIL 2019 Aim 1: To determine the potential for enhanced virulence and pathogenicity of Burkholderia cepacia complex (BCC) and Ralstonia strains isolated from the filtration unit of the ISS by genomic sequencing. Under Aim 1, to determine the potential for enhanced virulence and pathogenicity of BCC and Ralstonia strains isolated from the filtration unit of the ISS, the hypothesis follows that the sequencing and analysis of the 67 genomes of these two genera from 22 missions and their comparison to the genomic sequences of the respective ATCC BCC and Ralstonia strain controls will reveal known and novel pathogenicity islands and integrative conjugative elements (ICEs). The genes contained within these mobile genetic elements (MGEs) will allow us to gauge to what degree the presence of the genera in the ISS filtrated water systems pose a human health threat via their infection potential. Aim 1 objectives are complete; we have successfully sequenced all genomes and analyzed the dataset. We find that the ISS isolates have genomes that are distinct from that of terrestrial isolates and are set apart by the genes harbored on conjugative plasmids. Furthermore, among the ISS isolates plasmid composition also varies. We hypothesize that while the isolates exhibit varying degrees of genomic erosion, such erosion is mitigated by the presence of conjugative plasmids which can rescue a reduced genome when facing environmental stress. In the scenario of the ISS PWD, the isolates may form biofilms, as we have displayed in year 1 that Burkholderia species have the propensity to do so. Plasmids may be shuffled among the biofilm or aid planktonic isolates in overcoming the iodine challenges used to decontaminate the system. Year 2 transposon sequencing will shed light on whether or not this mechanism of plasmid shuffling is used to counter disinfection. Aim 2: To determine antibiotic persistence of BCC and Ralstonia strains isolated from the filtration unit of the ISS using Tnseq. Under Aim 2, performed in order to determine how the BCC isolates persist in the potable water system of the ISS, we first hypothesized that ISS strains, which have been placed under the selective pressure of the ISS water systems oligotrophic environment, will experience varying degrees of antibiotic tolerance due to this previous selective pressure. We found that their minimum inhibitory concentrations (MICs) do not differ from that of the terrestrial control under our antibiotics tested. However, a Tnseq analysis will reveal the mechanisms, including virulence factors that contribute to the ability of the BCC and Ralstonia strains to persist in the oligotrophic filtrated ISS water system. We hypothesize that genes conferred by conjugative plasmids aid in isolate survival of these bacterial species within the water system. The Year 2 goal for Aim 2 will include generating and assessing the gene survivorship index of a representative transposon library subject to distilled water conditions to simulate the environment of the ISS potable water system. In a Tnseq study, I will determine the genes associated with aiding in the maintenance of BCC bacteria in this ISS filtrated water system. Here a representative strain of BCC, which exhibits the greatest reduction in antibiotic susceptibility or tolerance will be selected for transposon mutagenesis. This technique of introducing a transposon to generate a library of insertional mutants and subsequent culturing of the library under a selective condition is known as Tnseq. The ISS specimen Tnseq library will be cultured in distilled water simulating the environmental conditions of the ISS water system. Subsequent sequencing analysis will allow us to determine which genes, including virulence factors, play a key role in the maintenance of the genera in the water systems of the ISS. All data will be made public and results will be published. Aim 2 objectives are 50% complete--we have successfully profiled all isolates for their antibiotic susceptibility. We are currently optimizing the Tnseq experiment. We expect to observe the genes which when mutated increase and detract from survival under distilled water treatment over time. We expect that the addition of iodine at 8 ppm or 40 ppm will alter this gene set in relation to the distilled water control. This comparison will reveal the mechanism used by the isolates to persist in the PWD despite disinfection treatment which is used to reduce the microbial load after stagnation in the ISS potable water system. We reason that if the genes conferred by plasmids are beneficial to overcome stress, then the mutants of these genes in the Tnseq library will be less represented in the iodine treated samples and therefore have a lower survival index than in the distilled water sample.
Bibliography Type:	Description: (Last Updated:)
Articles in Peer-reviewed Journals	O'Rourke A, Zoumplis A, Wilburn P, Lee MD, Lee Z, Vecina M, Mercader K. "Following the Astrobiology Roadmap: Origins, habitability and future exploration." Curr Issues Mol Biol. 2020 Jan 22;38:1-32. https://doi.org/10.21775/cimb.038.001 ; PubMed PMID: 31967574 , Jan-2020
Articles in Peer-reviewed Journals	O'Rourke A, Lee MD, Nierman WC, Dupont CL. "Genomic and phenotypic characterization of Burkholderia isolates from the potable water system of the International Space Station." PLoS One. 2020 Feb 19;15(2):e0227152. eCollection 2020. https://doi.org/10.1371/journal.pone.0227152 ; PubMed PMID: 32074104 , Feb-2020
Articles in Peer-reviewed Journals	Lee MD, O'Rourke A, Lorenzi H, Bebout BM, Dupont CL, Everroad RC. "Reference-guided metagenomics reveals genome-level evidence of potential microbial transmission from the ISS environment to an astronaut's microbiome." iScience. 2021 Feb 19;24(2):102114. https://doi.org/10.1016/j.isci.2021.102114 ; PMID: 33659879; PMCID: PMC7892915 , Feb-2021

Books/Book Chapters	O'Rourke A, Zoumplis A, Wilburn P, Lee MD, Lee Z, Vecina M, Mercader K. "Following the Astrobiology Roadmap: Origins, Habitability and Future Exploration." in "Astrobiology: Current, Evolving and Emerging Perspectives." Ed. A. Antunes. Caister Academic Press, 2020. p. 1-32. https://doi.org/10.21775/9781912530304 , Feb-2020